# GOWRISHANKAR S

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## PROFESSIONAL SUMMARY

Dynamic AI & Machine Learning Engineer specializing in Computer Vision and Deep Learning. Expert in developing and deploying production-ready models for real-time environments. Skilled in optimizing neural networks for edge devices and creating scalable data pipelines. Passionate about leveraging emerging technologies to solve complex industrial and real-world problems.

## TECHNICAL SKILLS

**AI & Deep Learning:** Machine Learning, NLP, Computer Vision (YOLOv8, OpenCV), PyTorch, TensorFlow, Transformers, Deep Learning.
**Engineering & Cloud:** Python, SQL, Node.js, Flask, FastAPI, Docker, Git/GitHub, Streamlit, Linux, API Integration
**Hardware & Data:** Raspberry Pi, IoT Sensor Integration, Power BI, Data Visualization, Google Sheets API

## PROFESSIONAL EXPERIENCE

**Forge Innovation and Ventures** July 2025 – Present, Coimbatore, Tamil Nadu
*Forge Fellow*

- Architecting and deploying production-ready AI/ML models specifically designed for low-latency, real-time data environments.
- Leading the development of end-to-end technology-driven products, from initial data acquisition to final cloud deployment.
- Implementing robust MLOps practices, including containerization with Docker and CI/CD pipelines for model versioning.

## KEY PROJECTS

**Visual Inspection System for Manufacturing**

- Developed a plug-and-play conveyor device utilizing 2D file comparison for automated quality assurance in manufacturing lines.
- Engineered computer vision algorithms to detect precise object dimensions and validate physical components against digital blueprints.
- Automated the validation process, significantly reducing manual inspection errors and increasing production throughput.
- Designed a seamless hardware-software interface for real-time defect identification and automated rejection mechanisms.

**AI-Powered Electronic Component Detector**

- Built an end-to-end detection system using Transformers and PyTorch to identify and classify electronic components in high-density PCB images.
- Implemented a Flask-based web interface supporting batch uploads, live camera streams, and confidence- based filtering for repair workflows.
- Optimized model inference for real-time video stream processing, providing instant feedback for assembly line operators.

**Object Navigation for Visually Impaired (Edge Computing)**

- Optimized heavy YOLO architectures for the Raspberry Pi 4 using quantization techniques, achieving real-time inference speeds.
- Integrated a multi-modal feedback system (Audio + Haptic) to provide spatial awareness and safe navigation paths.
- Designed a fail-safe navigation algorithm that prioritizes obstacle avoidance in dynamic indoor environments.